

Waveform-Processor (WP)

If the five waveforms provided by the FORMANT VCOs prove insufficient, the Waveform Processor offers the means—using relatively simple methods—to further modify these existing waveforms. Furthermore, the adjustable "Clipping" level is fully voltage-controllable. The user is thus provided with an (almost) infinite number of possibilities for variation!

The Waveform Processor (from the English *processing*—meaning manipulation or refinement) lives up to its name in the truest sense of the word. The circuit's design—which belongs to the category of signal-shaping circuits (Figure 1)—yields a multitude of new waveforms suitable for a range of interesting applications. Through "clipping" (cutting off the peaks of the waveform), through the (un)symmetrical amplification of both clipped and unclipped signal components, and through the inversion of the clipped waveform segments, it is possible—for instance—to brighten the darker timbres of sine and triangle waves. Likewise, effects such as the accentuation of the 3rd harmonic, frequency doubling, and the generation of trapezoidal waveforms are all possible. Furthermore, it is possible to voltage-control the "clipping level" (the inflection point) using LFOs, ADSRs, sequencers, etc. When driven by a slow sine or triangle wave (e.g., from an LFO), modulation frequencies between 0.5 Hz and 1.0 Hz produce phasing-like timbral dynamics; with slightly faster modulation (approx. 3 Hz to 10 Hz), these transition into chorus-like sounds. Sounds resembling those of a ring modulator can be generated by voltage-controlling a VCO oscillating within the audio range.

In both concept and design, the Waveform Processor is fully optimized for use within the FORMANT music synthesizer system. Consequently, the PCB layout was designed in the Eurocard format, even though there is ample space available (Figure 3).

During polyphonic playing, strong bass tones emerge as difference tones. Figure 2 illustrates—using four examples—some of the possibilities for "waveform synthesis" achievable with the Waveform Processor.

The unused space on the circuit board would have been more than sufficient to accommodate a second Waveform Processor or another circuit; however, this approach makes little sense if one intends to use a "compact" FORMANT front panel (Figure 4), where the numerous controls necessitate a densely packed layout (requiring 4 mm potentiometers). Constructing the Waveform Processor according to the component placement guide and parts list should be a straightforward process, particularly since no calibration is required.

Application Notes

Figure 5 illustrates the function of potentiometers P1 and P5. Finally, two "signal flow" diagrams are provided as a teaser for a musical and sonic journey of discovery.

Like the Ring Modulator, the Waveform Processor is a module designed for FORMANT users with an adventurous, experimental spirit.

Admittedly, some of the effects—a selection of which was already mentioned at the beginning of this description—could also be realized indirectly using the “conventional” FORMANT modules of the basic configuration. However, the construction of at least one Waveform Processor is highly recommended, not least due to its favorable price-performance ratio.

Reference: M. Bertuch: *Der Kurventransformator*, Patchcord, No. 3, p. 15.

Waveform Processor Bill of Materials

Resistors (Carbon Film, 5%):

R1, R2, R8, R9 = 56k

R3, R7 = 120k

R4 = 5.6k

R5 = 470k

R6 = 100k

R10 = 82k

R11 = 470Ω

Potentiometers:

P1 = 100k linear (Clipping Level)

P2 = 100k logarithmic (Clipping Mod.)

P3, P4 = 50k logarithmic (ES)

P5 = 10k linear (Symmetry)

Capacitors (MKH, MKS):

C1, C2, C3 = 560n (470n + 100n)

C4, C5 = 10μF/25V

Semiconductors:

D1 = DUS

IC1, IC2, IC3 = μA741C

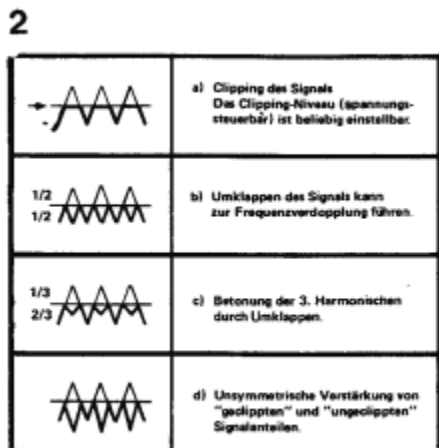
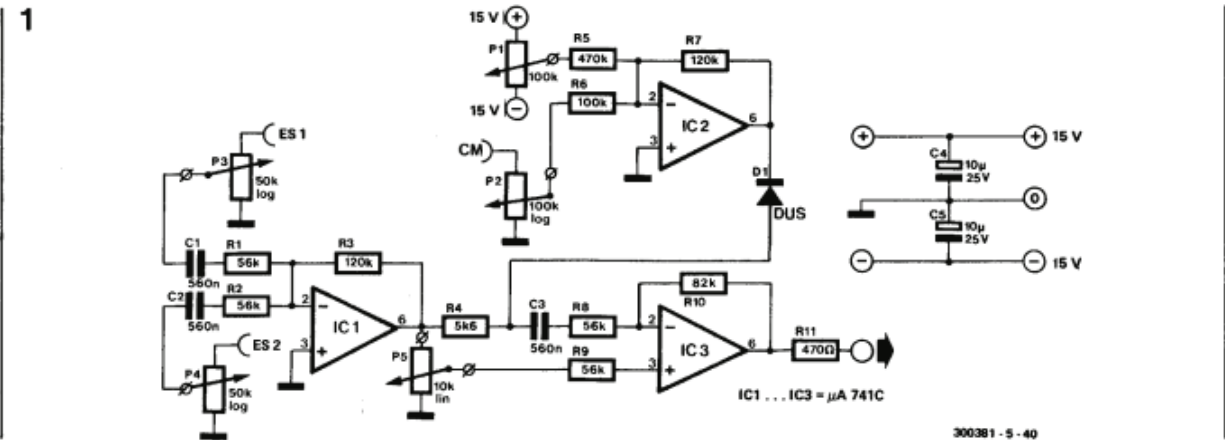
Miscellaneous:

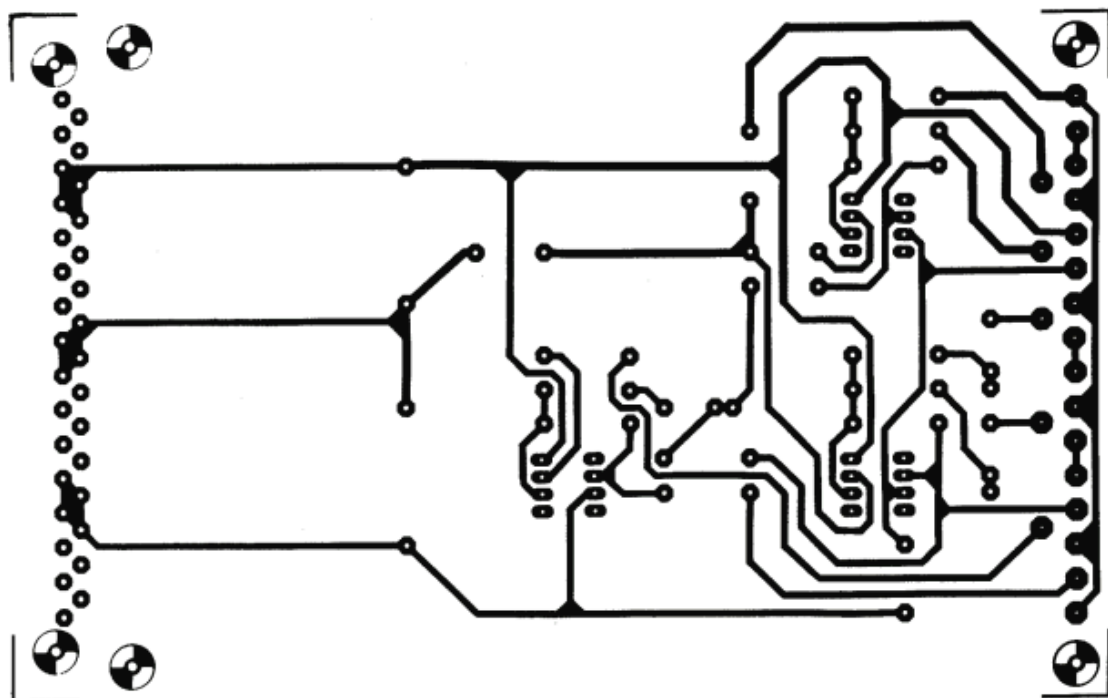
4 x 3.5 mm Jack Sockets

1 x 31-pin Header Strip or Solder Pins

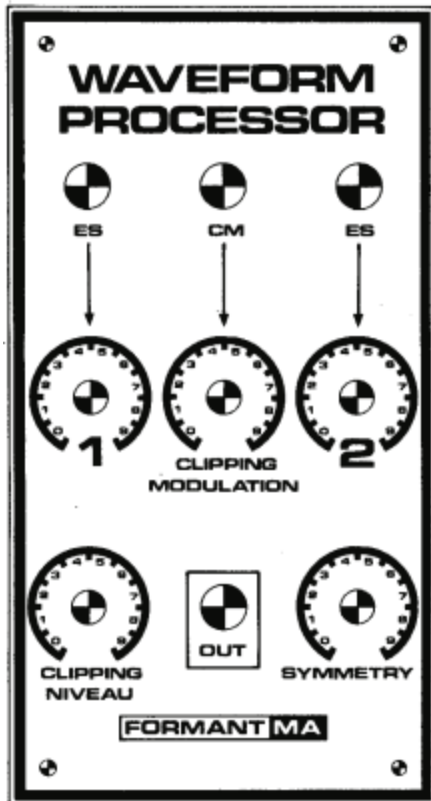
Figures

1. Schematic diagram for the Waveform Processor.
2. Waveform synthesis using the Waveform Processor (Input signal: Triangle wave).
3. PCB layout, component placement guide, and Bill of Materials for the Waveform Processor.
4. Suggested front panel design.
5. The effect of the "Clipping Level" (P1) and "Symmetry" (P5) potentiometers on a triangle wave signal.
6. Basic settings for a "clipping-modulated" sound.
7. Basic settings for a gong- or bell-like sound. Depending on the setting, the resonance filter emphasizes either the dark timbre of large percussion instruments or the bright timbre of small bells.

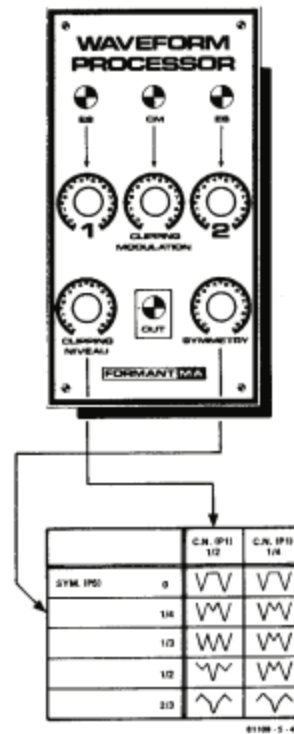




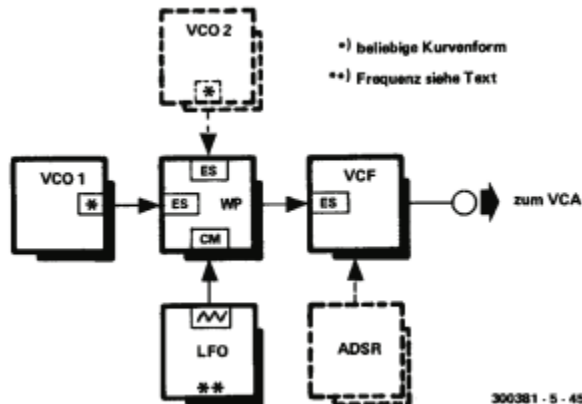
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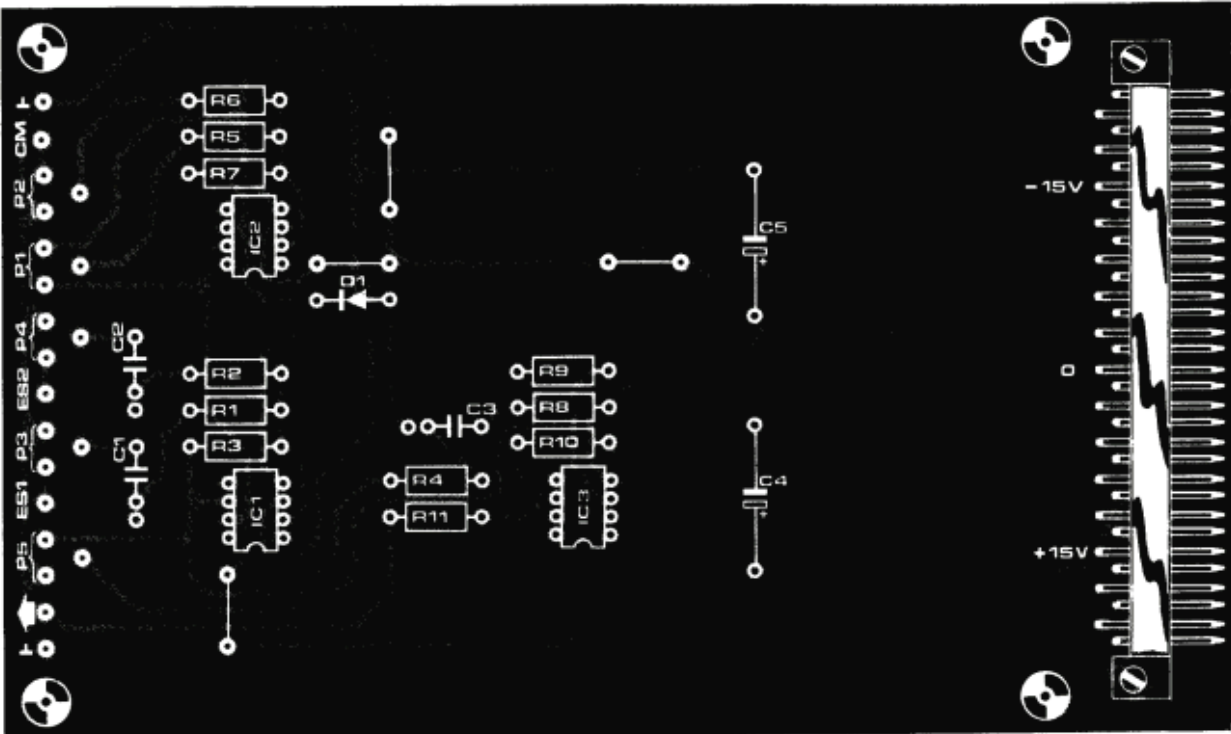


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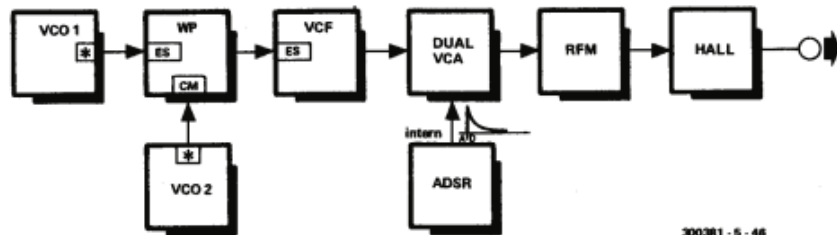


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*1) Kurvenform: Sinus, Dreieck oder Rechteck
Frequenz: der stärkste Effekt entsteht,
wenn die VCOs im Abstand einer Quart
(5 Halbtonen) gestimmt werden.